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XLII. *Rock Magnetism in India*

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Abstract

Measurements have been made of the magnetic polarization of 450 specimens of basaltic lavas of the Deccan Trap taken from two sites about 500 miles apart. With the exception of some nearly randomly magnetized specimens, the main direction of magnetization is N. 155° E. and the dip is 53° downwards. The most plausible interpretation of this result is (*a*) that India has drifted north from a position about 34° south of the equator when the rocks were formed some 70 million years ago and has rotated anti-clockwise through 25°, and (*b*) that either the earth's field was reversed when the rocks were formed or that the rocks became magnetized in the opposite direction to the field by some physico-chemical mechanism.

The pole position corresponding to these results does not agree with that found by other workers for British and American rocks of the same period. This suggests that a movement of India, relative to North America and Europe, has taken place at some time during the past 70 million years.

§ 1. Introduction

In October 1954 the Physics Department of Imperial College embarked on a three-year project of palaeomagnetic research in India. The scheme arose as a result of a previous survey made in Britain when it was found that certain sediments laid down over a wide area during the late Palaeozoic and early Mesozoic eras are consistently magnetized along a north–east south–west axis, and had a considerably lower inclination than the present earth's field (Clegg, Almond and Stubbs 1954). This result could only be explained with any degree of plausibility on the assumption that the land mass of England has drifted northward, and has rotated approximately 30° in a clockwise sense, relative to the earth's geographical axis, at some time since the end of the Triassic period. Further evidence of past movements of the British land mass has also been obtained from palaeomagnetic measurements made by Creer, Irving and Runcorn (1954), and the data now to hand are sufficiently

† Communicated by the Dr. J. A. Clegg.

conclusive to encourage the belief that by carrying out palaeomagnetic surveys on a global basis it may be possible to determine the positions occupied by the various land areas of the earth relative to the poles during different geological epochs. Recent work on American rocks by Du Bois (1955), Runcorn (1955) and Graham (1955) lends strong support to this expectation.

Among the areas of the world especially suitable for rock magnetic studies is the sub-continent of India. Of especial significance is the permo-carboniferous glaciation which suggests that the geographical latitude of India may not always have been what it is to-day.† Moreover, Indian geology has been studied in detail, and numerous exposures exist revealing a large part of the stratigraphical column. In particular, the Indian peninsular contains the important volcanic formation known as the Deccan Trap. This extensive series of basaltic lava flows covers an area of some 200 000 square miles, and was most likely extruded during the late Cretaceous and early Eocene, and so prior to the culmination of the Alpine orogeny when the great tectonic movements responsible for the raising of the Himalayas were in progress. Its advantages as a field for palaeomagnetic work were first pointed out by Irving (1954), who obtained a few samples of the Deccan Trap lavas from different localities, and found them to have a consistent axis of magnetization, directed roughly south in azimuth, but dipping towards the south. Despite the small number of specimens examined, the consistency of the results suggested that a more extensive magnetic survey of these rocks would be of interest, and we decided, in planning the present expedition, to concentrate initially on the Deccan Trap area.

The first phase of the work has now been completed. Two of us (E. R. D. and D. H. G.) spent two months in India during the winter of 1954–1955 and collected, in all, 208 rock samples. These were taken from a number of lava flows occurring in two different localities some 550 miles apart; the first in the neighbourhood of Linga near the District Capital of Chhindwara in the Central Provinces, and the second close to Khandala on the Bombay–Poona highway. The two areas will be considered separately.

The rock samples, which weighed in all about half a ton, were shipped to England for measurement. The standard method used for cutting cylindrical samples from the original blocks, and of determining the direction of magnetization using a simplified version of the astatic magnetometer described by Blackett (1952), has been described elsewhere (Clegg, Almond and Stubbs 1954).

2. Results from the Linga Area

Over an area of some 50 square miles around Linga there are a number of isolated exposures of basaltic lavas of the lower Deccan Trap series. The district has been mapped by Fermor and Fox (1916), who recognized

† For summary of some of the palaeo-climatological evidence see Holmes (1952) and Gutenberg (1951).

five separate flows which they designated by the numbers, 1, 2, 2*a*, 3 and 4 in ascending order. They estimated the age of these rocks to be late Cretaceous or possibly early Eocene and considered that the lowest flow (no. 1) might well be the oldest in the Deccan Trap series. A total of 69 samples was taken from nine sites including stream beds, quarries and road cuttings, spaced a few miles apart. At the first six sites to be visited it was possible to recognize each of the individual exposures as belonging to one of the flows 1, 2 or 2*a*. At the remaining three sites flows 3 and 4 were exposed, but it was not easy to differentiate between the two.

Laboratory measurements showed that all the samples with the exception of those from flow 2 have preferential directions of magnetization towards the south–east with rather steep downward dips. The combined results of the measurements of declination and dip for flows 1, 2*a*, 3 and 4 are shown as histograms in figs. 1 (*a*) and 1 (*b*), and on a Schmidt projection in fig. 3 (*a*).

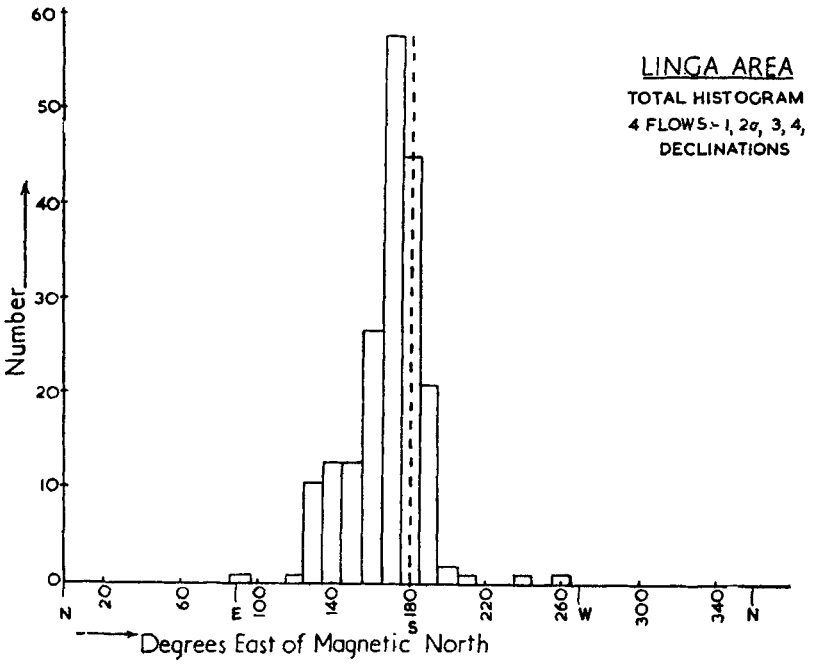
The samples from flow 2 were found to have random directions of magnetization, and the results have therefore been omitted from the histogram, but are included as open circles in the projection.

§ 3. Results from the Khandala Area

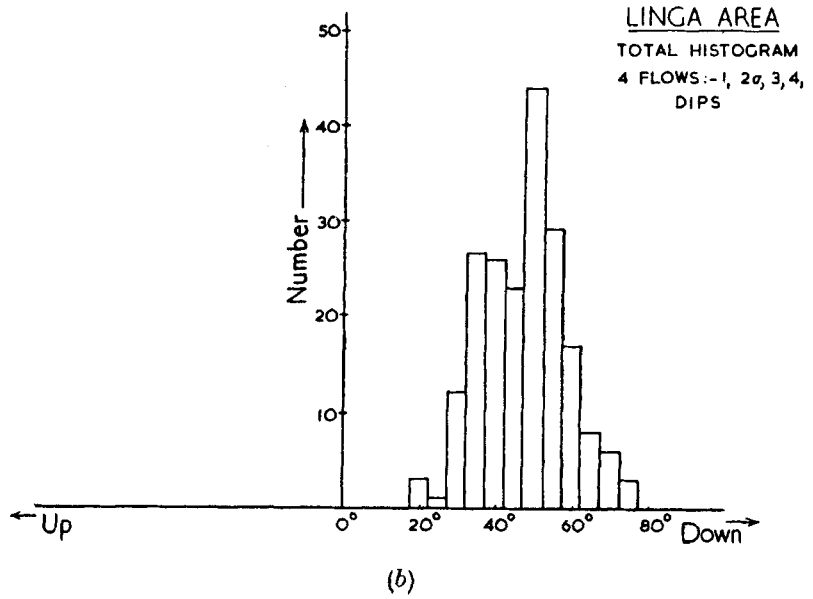
Close to Khandala, on the steep face of the Bhor Ghat, a large vertical sequence of basaltic lava flows are exposed in a relatively small area. They are thought to belong to the upper part of the middle section of the Deccan Trap, and according to Fermor and Fox (1916) they were probably laid down during the early Eocene period, and are younger than the Linga rocks. 139 rock samples were collected from 20 separate sites on the hillside, at heights ranging from 475 to 2525 feet above sea level. Since the individual exposures were on the whole small and scattered, and the flows numerous and often no more than a few feet in thickness, it was impossible to identify the latter individually.

The complete results for the whole Khandala area are shown in figs. 2 (*a*), 2 (*b*) and 3 (*b*). It can be seen from the histograms and the projection that the majority of the specimens are again magnetized in a general south–easterly direction with rather steep downward dips. A significant feature of the distribution in fig. 3 (*b*) is the group of points in the north–east quadrant and which produce the small secondary maximum in fig. 2 (*a*). They were all cut from four blocks, two of which came from the lowest flows at the base of the Ghat, and the other two of which were collected from the top of the mountain at heights of 2275 and 2525 feet. It will be seen from fig. 3 (*b*) that these anomalous specimens are magnetized in a direction considerably closer to the direction of the present earth's field than are the great majority of the specimens. It is therefore possible that their polarizations have been pulled round towards the present field, owing to partial magnetic instability. There are also three specimens from this area which show upward dips. These again were taken from a single clock belonging to one of the uppermost flows.

Fig. 1

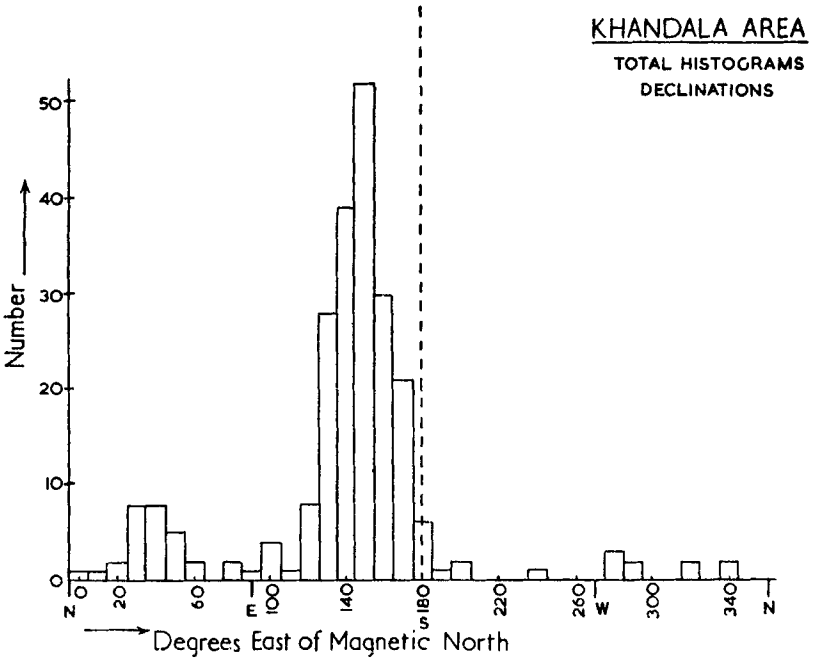

LINGA AREA
TOTAL HISTOGRAM
4 FLOWS:- 1, 2σ, 3, 4,
DECLINATIONS
Number
60
50
40
30
20
10
0
N
20
60
E
100
140
S
180
220
260
W
300
340
N
Degrees East of Magnetic North

(*a*)

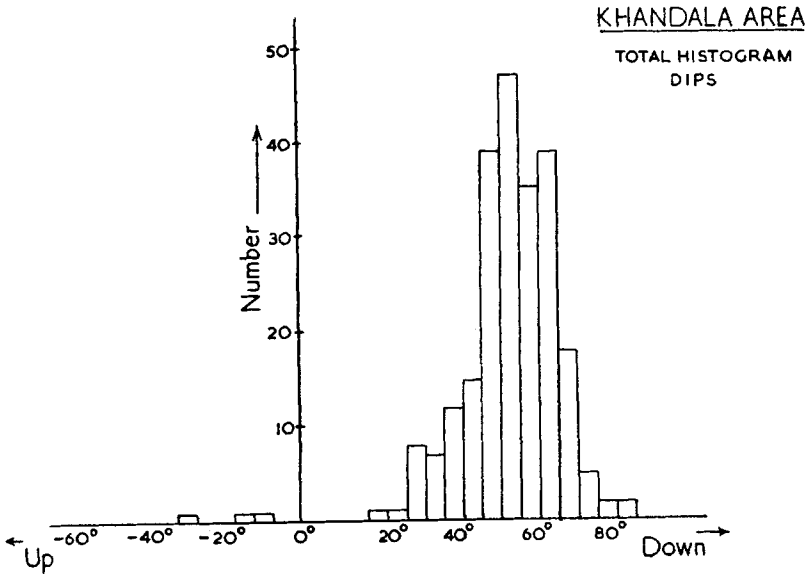

LINGA AREA
TOTAL HISTOGRAM
4 FLOWS:- 1, 2σ, 3, 4,
DIPS
Number
50
40
30
20
10
Up
0°
20°
40°
60°
80°
Down

(*b*)

Fig. 2


KHANDALA AREA
TOTAL HISTOGRAMS
DECLINATIONS
Number
50
40
30
20
10
0
N
0
20
60
E
100
140
180
S
220
260
W
300
340
N
Degrees East of Magnetic North

(*a*)


KHANDALA AREA
TOTAL HISTOGRAM
DIPS
Number
50
40
30
20
10
-60°
-40°
-20°
0°
20°
40°
60°
80°
Up
Down

(*b*)

1 Locality	2 Mean latitude	3 Mean longitude	4 Total no. of sites visited	5 Total no. of specimens measured *	6 Mean intensity of magnetization $\times 10^{-3}$ c.g.s.	7 Mean declination east of true south	8 Mean dip †	9 Radius of 50% circle of confidence ‡
Linga area	21° 58′ N.	78° 55′ E.	9	195§	$4{\cdot}72 \times 10^{-3}$	164°	+48° down	1·0°
Khandala area	18° 45′ N.	73° 22′ E.	20	233	$3{\cdot}12 \times 10^{-3}$	139° (147°)¶	+62° down (+58°)¶	1·4°

Mean for both sites N. 155° E. and +53° down.

* The term ' specimen ' applies to the individual cylinders cut from the original blocks. Two or more of these were often taken from a single block.

† The dips were measured relative to a horizontal plane.

‡ Computed by the method of Fisher (1953).

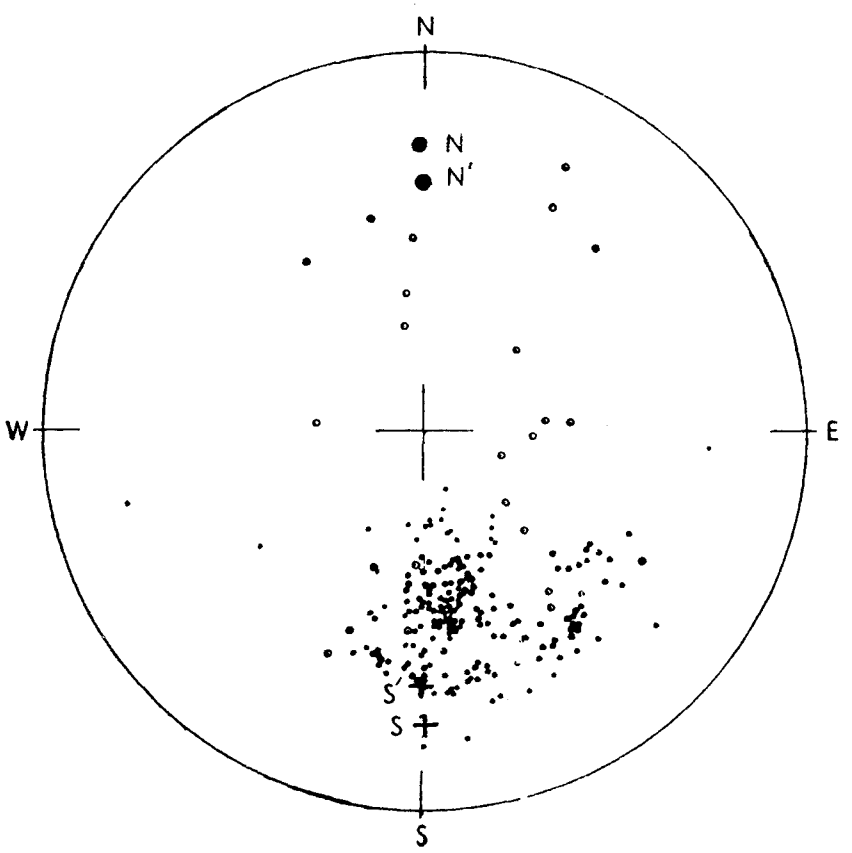
§ These specimens were taken from flows 1, 2*a*, 3 and 4. Flow 2 was randomly polarized and the results have not been included.

¶ The figures in brackets give the mean declination and dip for the two sites when the anomalous specimens from the Khandala area are omitted.

§ 4. Discussion of Results

Further details of the results for the two areas are given in the table. Although the declinations of the individual samples were measured with respect to magnetic north, the mean values shown in column 7 of the table

Fig. 3



(*a*)

Linga Area—Flows 1, 2*a*, 3 and 4: ●; Flow 2: ○.

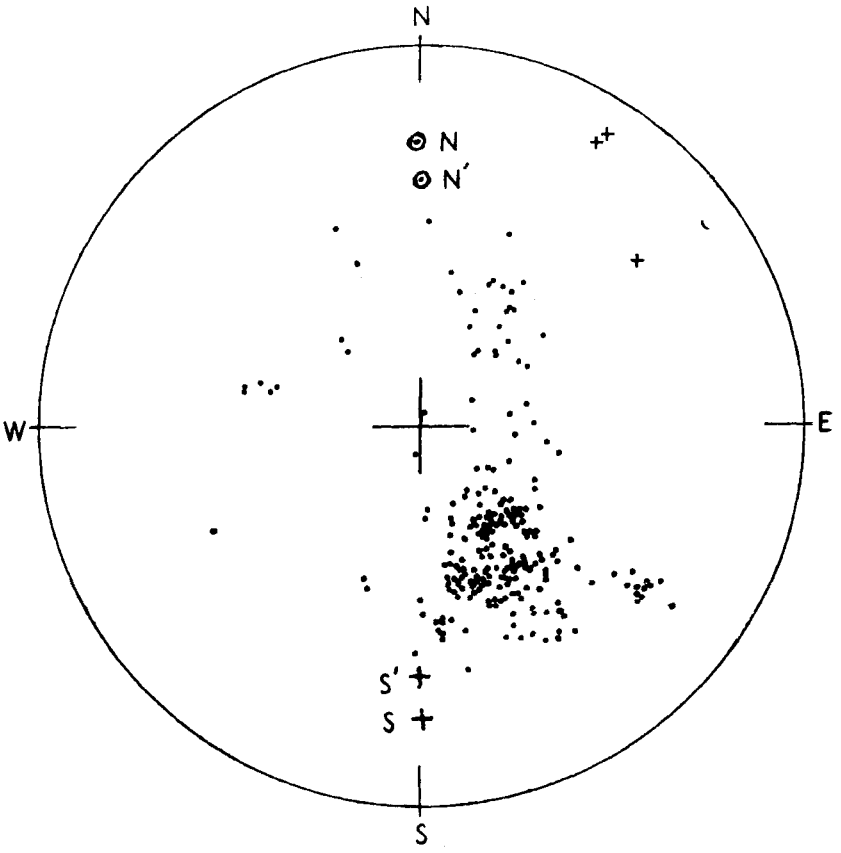
Schmidt Projection

Circles and dots, lower hemisphere: ● ○; Crosses, upper hemisphere: + ; North Pole, present: ● N; dipole: ● N′; South Pole, present: + S; dipole: + S′.

are expressed in degrees east of true north. In the particular region concerned the declination is very small, being 0·6° west. The lava flows sampled were all very nearly horizontal, and the magnetic dips (column 8) were therefore measured relative to a horizontal plane. This marked

horizontality over large areas is a characteristic feature of the Deccan Trap lavas, which are believed to have been laid down as level flows and to have remained free, for the most part, from subsequent folding or tilting (Fermor and Fox 1916).

Fig. 3



(*b*)

Khandala Area—All flows : ●.

Schmidt Projection

Circles and dots, lower hemisphere : ● ○ ; Crosses, upper hemisphere : + ; North Pole, present : ● N ; dipole : ● N′ ; South Pole, present : + S : dipole : + S′.

The figures given in the table show that the general direction of magnetization is similar in both areas. However, the rocks from Khandala have, on the whole, a significantly bigger downward dip and a more easterly declination than those from Linga. This difference is partly

due to the anomalous and probable partially unstable specimens from the Khandala, which have already been discussed. If these 23 anomalous specimens are omitted, the mean values for declination and dip for Khandala obtained from the remaining 210 specimens become 147° and +58° (shown in brackets in the table), which are somewhat closer to Linga figures.

The origin of the random scatter of the specimens from flow 2 in the Linga area is unknown. Lightning, weathering or some other disturbance are possible causes.

Though the difference between the results from the two sites is still significant and is under further study, for the purposes of the present discussion we will use the mean value for the two locations. Thus we get for the direction of the magnetic intensity (north seeking pole) when the rocks were magnetized, the following figures :

Declination	N. 155° E.
Inclination	+53° down.

It will be seen from figs. 3 (*a*) and (*b*) that the direction of magnetization of most of the rocks is almost perpendicular to that of the present magnetic field in India. The majority of the rocks, with the possible exceptions which have already been mentioned, must therefore have had a considerable degree of magnetic stability over long periods of time, so that it is likely that their polarizations give a true indication of the earth's field at the time of cooling. The relatively narrow peaks in both declination and dip shown in figs. 1 (*b*) and 2 (*b*) strongly suggest a high degree of stability for most of the specimens. For when magnetic instability in rocks is found it generally affects different specimens differently, leading to a wide distribution in measured directions.

We conclude that throughout the whole period during which the lava flows of the Linga and Khandala were being laid down, the earth's magnetic field was directed either along a south-east axis dipping downwards, or, if the rocks were self-reversed, along a north-west direction dipping upwards. Since the period of time over which the Deccan Trap was extruded was probably much longer than the period of the secular variation, we can reasonably assume that the mean direction of magnetization of all the measured flows should be close to that corresponding to an axial dipole, which we assume to be co-axial with the geographical axis of the earth.

Our results therefore suggest that the Indian peninsular occupied a very different position relative to the geographical poles during the late Cretaceous and early Eocene periods from that at the present time. In particular, in order to account for the rather steep downward dip of 55° compared† with the present 25°, it is necessary to suppose that India

† It should be noted that the dip in India corresponding to an axial dipole is 36°.

was then situated considerably farther from the equator. For if the earth's field corresponded on the average to that of an axial dipole, the relationship between the mean magnetic dip, θ, and the geographical latitude, λ, would be given by

$$\tan \lambda = \tfrac{1}{2} \tan \theta.$$

Substituting the mean values of θ of 53° from column 8 of the table, we obtain the values of 34° for λ. After taking into account the direction and sign of the dips, we therefore have two possibilities, namely :

(*a*) This particular area of Central India may then have been about 34° south of the equator. In this case it is necessary to assume either that the direction of the field was opposite to that at the present time, or alternatively that the rocks took on a reverse magnetization on cooling. We must therefore suppose that the whole sub-continent has moved northward across the equator at some time during the intervening 70 million years between that time and the present day. The south–easterly declinations indicate that this transitional movement was accompanied by an anti-clockwise rotation of 25°.

(*b*) The observed directions of polarization are also compatible with the assumption that the rocks were formed in the northern hemisphere, at a position approximately 34° north of the equator. But if this were so, the directions of the declinations and dips show that India must have rotated through nearly 180° or alternatively have drifted across one of the poles to its present position.

Of these various possibilities the first, that of a northward drift and a small clockwise rotation, is evidently by far the most plausible both on physical and geological grounds.

We thus finally conclude that the results are best explained on the postulate that India lay about 34° south of the equator during the late Secondary and early Tertiary eras, and that at some subsequent time it moved into the northern hemisphere and rotated about 25° clockwise relative to the earth's geographical axis. A latitude shift from 34° south to the present 20° north, that is, of 54°, corresponds to a linear movement of 5000 km.

It is very hard to see how this conclusion can be avoided except by some very special and unlikely hypothesis. A conceivable way of reconciling the polarization of these rocks with the present geographical position of India would be to assume that the polarizations were all originally along the reversed direction of the present field (i.e. south and 25° upwards, point S in figs. 3 (*a*) and (*b*) and that they have been just sufficiently unstable to have been all pulled round through some 90° to the present position of the peak, that is, 25° east and south and 55° downwards. This is so unlikely that it will be provisionally rejected.

In considering the mechanism whereby such a large movement of India might have occurred, one is immediately tempted to consider tectonic causes, since current theory of mountain building suggests that

India evidently advanced towards the main Asiatic land mass during the building of the Himalayas.

Now the extent of this movement of India, relative to Siberia during the Alpine orogeny, is in great doubt and various authorities give figures varying from 100 km (Jeffreys 1906) to 3000 km (Van der Gracht 1928). Taking a few hundred kilometres as a reasonable mean estimate of the relative motion of India and Siberia, the rock magnetic measurements would indicate a much bigger northward movement of India relative to the geographical axis than relative to Siberia. Whether, in fact, Siberia shared in this greater northward movement or not, should be determinable by rock magnetic measurements in Siberia.

§ 5. Five Comparisons with Other Work

It has been shown by Runcorn (1955) and more recently by Graham (1955) that the directions of magnetization found in certain British, North American and Icelandic rocks are consistent with the assumption that the earth's mantle has moved as a whole—that is, according to the polar wandering hypothesis.

Of particular significance, by reason of the large number of specimens studied, is Graham's comparison of his results for Permo-Triassic sandstones in North America with the results of Clegg *et al.* for the Triassic sandstones of England. Taken together these results indicate a pole position 180 million years ago as 45° N. and 130° E.

Graham considers that no rotation or translation of Europe relative to North America as great as 20° would be consistent with the magnetic results. Gold (1955) has given theoretical arguments which indicate that polar wandering is much more likely to have occurred than relative motion of continents.

It is interesting to consider our results for the Deccan Trap in relation to these hypotheses. The position of the north pole in Eocene-Cretaceous times (70 million years ago) as deduced from our measurement is

28° N. 85° W.

This can be compared with the estimate by Runcorn of the pole position in Eocene-Cretaceous times, as deduced from measurements of British, North American and Icelandic rocks, which is

76° N. 130° E.

It is evident that this large difference between the pole position calculated from our results and that calculated by Runcorn indicates that India must have moved relatively to Northern Europe and North America.

Some slight evidence for a rotation of India relative to South Africa is obtained by comparing our results with those of Gough (1955) for the Pilansburg Dyke system, which is thought to be between 300 and 400

million years old. These measurements gave the following values for the direction of the polarization :

Declination N. 24° E.
Inclination +69° down.

The corresponding position of the pole—the north pole if the earth's field was reversed or the south pole if it was not—is given by Gough as 8° north and 43° east, which is situated in Abyssinia.

Using similar arguments to those which we have applied to our Indian results, one can conclude that the most likely explanation of Gough's result is that South Africa has drifted right across the south pole to reach its present position and has rotated 24° clockwise in so doing. Although the general northward movement of South Africa in the last 300 to 400 million years is in the same direction as that of India in the last 70 million years, the relative rotations are in the opposite sense : for South Africa 24° clockwise, for India 25° anti-clockwise ; this gives our relative angular rotation of nearly 50°. However, the wide differences in ages of the rocks makes this comparison of only suggestive importance.

Whatever the mechanism responsible for the postulated northward drift of the Indian peninsular, our present results clearly indicate many lines of future research in rock magnetic studies in India.

In the first place, the discrepancy between the measurements at Linga and Khandala require further investigation. It seems possible that some part of the difference may be due to the secular variation. To investigate this more fully, it is evidently necessary to carry out a more detailed survey of the two locations, and to extend the scope of the work to cover other parts of the Deccan Trap area. It will be noted that the Linga rocks have the smaller dip ; whereas if they are in fact older, they would have been expected to have larger dips, due to the postulated northward drift and so to their greater southern latitude when formed.

Secondly, it is well known that Southern India suffered heavy glaciation during Permo-Carboniferous times. If climatic conditions then prevailing over the earth were similar to those of the present day, this would suggest that some 250 million years ago India was in a high southern latitude, say, not less than 65° south. The latitude of 34° south found in the present investigations for Cretaceous-Eocene times is intermediate between this and the latitude today. It is therefore important to make further measurements on rocks of both pre-Cretaceous and post-Eocene ages in order to obtain more detailed information concerning the position of India during these periods. This investigation is now proceeding.

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